

CURRICULUM VITAE

1. PERSONAL DATA

Name: Muhammad Sulaiman

Place of Birth: Buner-Khyber Pakhtunkhwa

Nationality: Pakistan

Contact Address: House 510, Street 16, Sector S, Sheikh Maltoon,
23200 Mardan, Khyber Pakhtunkhwa, Pakistan.

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Email: msulaiman@awkum.edu.pk, sulaiman513@gmail.com

Spoken language skills: English (Advanced), Urdu (Advanced), Pashto (Advanced)

2. ACADEMIC/PROFESSIONAL PARTICULARS

(a) Field of Specialization:

Applied and Computational Mathematics; Global and Engineering Optimization; Evolutionary Computation and Nature-Inspired Metaheuristics; Scientific Machine Learning; Physics-Informed Neural Networks; Numerical Differential Equations; and Mathematical Modelling.

(b) Academic Qualifications: (DD/MM/YYYY)

PhD, Mathematics, 25/10/2011--07/04/2015, University of Essex, United Kingdom, A Nature-inspired Metaheuristic: The Plant Propagation Algorithm, Passed with no corrections.

M Phil [Two Years], Mathematics, 15/01/2007--19/02/2009, Quaid-i-Azam University Islamabad, Pakistan, Extension of Noether's theorem to constrained non-conservative dynamical systems in Poincaré formalism, First Division with 77%.

M Sc [Two Years], Mathematics, 03/01/2005--05/01/2007, Quaid-i-Azam University Islamabad, Pakistan, First Division with 70%.

BSc [Two Years], 01/10/2002--09/10/2004, Mathematics-A/ Mathematics-B/ Computer Science, University of Peshawar, Pakistan, First Division with 72%.

Higher Secondary School Certificate (FSc) [Two Years], 01/08/2000--31/08/2002, Mathematics/Physics/Chemistry, BISE Mardan, Pakistan, First Division with 70%.

Secondary School Certificate (SSC), 02/03/1998--03/07/2000, Mathematics/Physics/Chemistry/Biology, BISE SWAT, Pakistan, First Division with 70%.

(c) Academic Honors and Awards

- Awarded Faculty Development Scholarship, Government of Khyber Pakhtunkhwa, Pakistan, 2011

- Silver medal for best paper in Engineering Design, INNOVATION & TECHNOLOGY TRANSFORMATION POST-COVID-19, Universiti Sultan Zainal Abidin (UniSZA), Malaysia, 2022

(d) Membership of Professional Bodies

- Higher Education Academy United Kingdom, Associate Fellow, 2014
- Higher Education Commission Pakistan, Approved PhD Supervisor, 2015-till date

(e) Language Proficiency

English spoken/written advanced level of proficiency.

Urdu spoken/written advanced level of proficiency.

Pashto spoken/written advanced level of proficiency.

(f) Software Skills

Sl. No.	Skill Title	Skill Level (Basic/Intermediate/Proficient)	Total Experience (years)	Last Used (Year)
1	MATLAB	Proficient	9	2026
2	Python	Proficient	9	2026
3	MS Office	Proficient	9	2026
4	LaTeX	Proficient	9	2026
5	Origin Lab	Proficient		
6	Online Teaching tools (Zoom, MS Teams, Team Viewer, RustDesk)	Proficient	9	2026
7	Mathematica	Proficient	9	2026
8	Maple	Proficient	9	2026

3. CAREER DETAILS

(a) Academic Positions Held (DD/MM/YYYY)

Full Professor (Tenured), Department of Mathematics, Faculty of Physical and Numerical Sciences, Abdul Wali Khan University Mardan, Pakistan,

Period (13/09/2025-till date)

Associate Professor (Tenured), Department of Mathematics, Faculty of Physical and Numerical Sciences, Abdul Wali Khan University Mardan, Pakistan,

Period (13/08/2021-12/09/2025)

Assistant Professor (Tenure Track), Department of Mathematics, Faculty of Physical and Numerical Sciences, Abdul Wali Khan University Mardan, Pakistan,

Period (22/02/2016-12/08/2021)

Lecturer (BPS-18), Department of Mathematics, Faculty of Physical and Numerical Sciences, Abdul Wali Khan University Mardan, Pakistan,

Period (19/08/2009-21/02/2016)

4. Teaching

(a) Summary of Courses Taught (undergraduate and postgraduate)

- Bio-inspired Algorithms for Optimization
- Advanced Metaheuristics for Optimization
- Evolutionary Algorithms and Applications
- Neural Networks and Optimization Algorithms
- Optimization Theory and Computational Methods
- Introduction to Optimization Theory
- Engineering Optimization
- Ordinary Differential Equations
- Partial Differential Equations
- Calculus-I, II
- Linear Algebra-I
- Real Analysis
- Basic Mathematics
- Introduction to Artificial Intelligence
- Programming for Artificial Intelligence

(b) Participation in Academic Accreditation

- Higher Education Academy United Kingdom, Associate Fellow,
Period (2014)

(c) Research Students Supervised/Trained

- PhD graduates supervised to completion as principal supervisor: 8
- MPhil graduates supervised to completion as principal supervisor: 40
- BS Mathematics research projects supervised: 38

(d) Participation in Thesis and Oral Examination Committees

- (1) University of Malakand, Pakistan (BS, MPhil and PhD Examiner)
- (2) HITEC University Taxila, Pakistan (MPhil Examiner)
- (3) Kohat University of Science and Technology, Pakistan (MPhil and PhD Examiner)
- (4) Shaheed Benazir Bhutto Women University Peshawar, Pakistan (PhD Examiner)
- (5) University of Swat, Pakistan (MPhil Thesis Examiner)
- (6) The University of Engineering & Technology, Peshawar, Pakistan (MPhil Examiner)
- (7) COMSATS University Islamabad (MPhil Thesis Examiner)

(e) Training

- Completed the University of Essex, UK, Induction Program for Graduate Teaching Assistants
- Equality & Diversity Essentials course with a score of 88 % at the University of Essex UK

5. RESEARCH

(a) Research Interests

Applied and Computational Mathematics; Global and Multi-objective Optimization; Engineering Design Optimization; Evolutionary Computation and Nature-Inspired Metaheuristics; Scientific Machine Learning and Physics-Informed Neural Networks; Artificial Neural Networks and Hybrid Neuro-Optimization Methods; Numerical Solutions of Ordinary, Partial, and Fractional Differential Equations; Mathematical Modelling of Complex Physical, Biological, Environmental, and Cyber Systems; Fluid Dynamics, Heat and Mass Transfer; Soft Computing and Computational Intelligence.

(b) Publications/Citations Data

Post PhD experience by 12 July 2026	11+ years
Total Teaching and Research Experience	16+ years
Major and specialization/concentration	Applied Mathematics, Optimization Theory, Computational Mathematics, Machine Learning
H-index	30 (Google Scholar)
total citations	2693 (Google Scholar, 12 July 2026)
Google scholar link	https://scholar.google.com/citations?user=Ikq95ogAAAAJ&hl=en
ResearchGate Link	https://www.researchgate.net/profile/Muhammad-Sulaiman-23
LinkedIn Profile	https://www.linkedin.com/in/muhammad-sulaiman-mathematics/
Web of Science Profile	https://www.webofscience.com/wos/author/record/360390
Scopus Profile	https://www.scopus.com/authid/detail.uri?authorId=57212956089
ORCID	0000-0002-4040-6211
The number of publications in Q1 journals.	30
The number of publications in Q2 journals.	48
The number of publications in Q3 journals.	6
The number of publications in Q4 journals.	0
Cumulative Impact Factor	285+
The number of publications in non-indexed journals.	19
The number of published books.	N/A
The number of published <i>chapters of books</i> .	2
The number of Graduate Students supervised and the role.	PhD: 08, MPhil: 40

(c) Research Grants

- Completed a research project titled: "Application of Soft Computing Techniques for Solving Real-life Problems" as **Principal Investigator** in collaboration with Universidad Santiago de Cali, Cali 76001, Colombia, 2021.
Project Amount: USD 26008
- Higher Education Pakistan, Startup Research Grant 2016. As **Principal Investigator**.
Amount: PKR 0.2 million.

- Research grant provided by The Deanship of Scientific Research (DSR), Vice Presidency for Graduate Studies and Scientific Research, King Faisal University (KFU), Ministry of Education, Saudi Arabia, 2022. Amount: USD 6000
- Ref: FRGS/1/2022/STG06/UMT/03/3 : 2022-2025

Working as a **Foreign Researcher**. Project Amount: USD 15900.

"A new neuro soft-computing based on green former matrix factorization of high dimensional partial differential equations for ecological modelling", Fundamental Research Grant Scheme (FRGS), Ministry of Higher Education, Malaysia.

- AWARD OF UNDERGRADUATE RESEARCH SUPPORT PROGRAM GRANT UNDER THE PROJECT BUILDING PROVINCIAL STI SYSTEM, **As Supervisor**. [Rs 150000/-] The Directorate General of Science & Technology Khyber Pakhtunkhwa, research titled "A machine learning framework for evaluation of COVID-19 model with asymptomatic and symptomatic profiles infected individuals." [2022-2023]

(d) Participation in Regional & International Conferences

- 1) EGH2012, Essex-Greenwich-Hertfordshire Workshop on Applied and Numerical Mathematics (sponsored by the London Mathematical Society), June 7th – 8th 2012, University of Greenwich, Greenwich Campus, Park Row, Greenwich, London.
- 2) International Conference on Metaheuristics and Nature Inspired Computing, META'2014, October 27th – 31st 2014, Marrakech, **Morocco**.
- 3) Soft Computing and its Practical Applications, 25-26 Oct 2016, Kohat University of Science and Technology, Kohat.
- 4) Modelling Simulation and Optimization, 18-19 April 2017, KUST
- 5) NUST Conference on Recent Trends in Mathematical Sciences, 7-8 Nov 2017, National University of Science and Technology (NUST), School of Natural Sciences (SNS), Islamabad, Pakistan.
- 6) The Role of Mathematics in Solving Real-World Problems, 29-30 Nov 2017, KUST International Conference on Emerging Technologies (ICET), 27-28 DEC 2017, Capital University of Science and Technology, Islamabad.
- 7) The first international Metaheuristics Summer School – MESS 2018, 21-25 July 2018, Acireale, **Italy**.
- 8) Emerging Trends in Scientific Computing, March 28-29th, 2019, Kohat University of Science and Technology, Kohat, KP, Pakistan.
- 9) International Conference on Mathematics, 1st OUL-ICM 2019, Nov 30-Dec 1 and 2 2019, Lahore, Pakistan.
- 10) The NUST Conference on Recent Trends in Mathematical Sciences-II, NUST, Dec 3 and 4, 2019, Islamabad, Pakistan.
- 11) The first Malakand-Japan Conference on Mathematics and its Application, March 3-5, 2020 at University of Malakand, Pakistan.
- 12) Online Webinar on "Intelligent Optimization Techniques and their real-life applications" organized by Punjab University, Patiala. [\[Certificate\]](#)
- 13) 2nd International Conference on "Orthogonal Polynomials, Special functions and

- Computer Algebra: Applications in Engineering" (OPSFCA-2022)
on October 15th – 16th, 2022. [\[Certificate\]](#)
- 14) 6th International Conference on Multidisciplinary Design Optimization and Applications,
Paris, France, **(Online mode)** October,16-19, 2024, [\[Link to conference site\]](#)

6. SERVICE

(a) Membership of Conference Committees

- 1) TPC Member for "2016 12th International Conference on Natural Computation, Fuzzy Systems and Knowledge Discovery (ICNC-FSKD 2016)"
- 2) TPC Member for "2017, 2018- 13th and 14th International Conference on Natural Computation, Fuzzy Systems and Knowledge Discovery (ICNC-FSKD 2017-2018) [\[Link\]](#)
- 3) Member of Scientific Committee: RECENT ADVANCES IN PURE AND APPLIED MATHEMATICS 20-22 January 2016 [\[Link\]](#)
- 4) Member of Scientific Committee: 3rd International Conference on Advanced Engineering Technologies (ICADET), Bayburt University Turkey, on 19-21 September 2019. [\[Link\]](#)
- 5) Program Committee Member of 14th Metaheuristics International Conference 11-14 July 2022, Ortigia-Syracuse, Italy [\[Link\]](#)

(b) Service as Reviewer

- 1) Journal of the Operational Research Society
- 2) International Journal of Computing Science and Mathematics
- 3) The journal OPSEARCH
- 4) Neural Computing and Applications
- 5) RAIRO - Operations Research
- 6) The Nucleus, PINSTECH, Nilore, 45650, Islamabad, Pakistan
- 7) International Journal of Metaheuristics
- 8) Independent Journal of Management and Production
- 9) Sindh University Research Journal (SURJ)
- 10) IEEE ACCESS
- 11) Mathematics
- 12) Energies
- 13) Sustainability
- 14) Transactions on Intelligent Systems and Technology
- 15) Fractal and Fractional
- 16) Universe
- 17) International Journal of Bio-Inspired Computation (IJBIC)
- 18) International Journal of Swarm Intelligence Research (IJSIR)
- 19) Journal of Intelligent and Fuzzy Systems
- 20) Journal of Computational Design and Engineering
- 21) IEEE Transactions on Systems, Man and Cybernetics: Systems
- 22) Reviewer for Edited Book "Big Data Analytics in Supply Chain Management" [\[Link\]](#)
- 23) International Journal of Information Technology & Decision Making
- 24) PLOS ONE
- 25) Journal of Mathematics
- 26) Soft Computing

(c) Membership of Journal Editorial Boards

- 1) COJ Reviews and Research [\[Link\]](#)
- 2) SCIREA Journal of Mathematics [\[Link\]](#)
- 3) Annals of Applied Sciences [\[Link\]](#) [Indexed: CrossRef, Google Scholar]
- 4) Innovative Journal of Mathematics [\[Link\]](#)
- 5) **[Special Issue]** Energies (MDPI) [\[Link\]](#)
- 6) **[Special Issue]** Computers and Electrical Engineering (Elsevier) [\[Link\]](#)
- 7) **[Special Issue]** IEEE Journal of Biomedical and Health Informatics [\[Link\]](#)
- 8) **[Special Issue]** Fuzzy Information and Engineering, ISSN 1616-8658 e-ISSN 1616-8666 [\[Link\]](#)
- 9) **[Special Issue]** Computing and Informatics, [\[Link\]](#)

(d) University Service

Abdul Wali Khan University Mardan, Pakistan, Period (04/15-till date)

- Member of Board of Studies
- Convener of Seminar Organizing Committee
- Focal Person of Teacher Evaluation Team
- Member of Graduate Studies Committee
- Member of Departmental Tenure Review Committee
- Member of Scholarship Committee
- Member of Curriculum and Syllabi Review Committees for Standardization & Uniformization of BS Courses Across HEIs/DAIs of Khyber Pakhtunkhwa

7. List of Publications

Link to publications:

https://scholar.google.com/citations?hl=en&user=Ikq95ogAAAAJ&view_op=list_works&sortby=pubdate

(a) International Refereed Journals (Published only!)

(JCR 2026) Quartile	Number of Papers Published
Q1	30
Q2	48
Q3	6
Q4	0
Book Chapters	2
Impact factor	285+

2026

[107] Rahman, Irfan Ur; **Sulaiman, Muhammad**; Alshammari, Fahad Sameer; Laouini, Ghaylen. “Numerical study of nonlinear flow and mass transfer through a channel filled with a porous medium using a hybrid heuristic paradigm.” *Waves in Random and Complex Media*, Vol. 36, No. 2, pp. 1448–1483, 2026. ISSN: 1745-5030. Publisher: Taylor & Francis. Reference type: Journal Article.

[106] **Sulaiman, Muhammad**; Fazal, Fazlullah; Alshammari, Fahad Sameer; Laouini, Ghaylen. “Dynamics of bioconvection radiative MHD flow with gyrotactic microorganisms: an intelligent computation approach.” *Waves in Random and Complex Media*, Vol. 36, No.

3, pp. 3874–3906, 2026. ISSN: 1745-5030. Publisher: Taylor & Francis. Reference type: Journal Article.

[105] **Sulaiman, Muhammad**; Hussain, Zawar; Alshammari, Fahad Sameer; Laouini, Ghaylen. “Computational Analysis of Natural Convection Heat Transfer in Nanofluids Under a Uniform Magnetic Field Using Levenberg–Marquardt Backpropagation Neural Networks.” *Journal of Computational Applied Mechanics*, Vol. 57, No. 1, pp. 41–62, 2026. ISSN: 2423-6713. Publisher: University of Tehran Press. Reference type: Journal Article.

[104] Saad, Muhammad; **Sulaiman, Muhammad**; Khan, Muhammad Fawad; Laouini, Ghaylen; Ashraf, Rashid; Alshammari, Fahad Sameer. “Magnetic Hydrodynamic Flow and Heat Transfer of Williamson Nanofluids in a Porous Medium Impact of Chemical Reactions and Melting Effects.” *Journal of Computational Applied Mechanics*, Vol. 57, No. 2, pp. 173–194, 2026. ISSN: 2423-6713. Publisher: University of Tehran Press. Reference type: Journal Article.

[103] Jan, Hamad; **Sulaiman, Muhammad**; Khan, Naveed Ahmad; Alshammari, Fahad Sameer; Laouini, Ghaylen; Ashraf, Rashid; Ali, and Mustafa Ahmed. “Improved pipeline damage detection using flexural and torsional ultrasonic guided waves.” *Research in Mathematics*, Vol. 13, No. 1, 2026. Publisher: Taylor & Francis. Reference type: Journal Article.

[102] Alkathlan, Ali; Kateb, Faris; Ahmad, Tauqeer; **Sulaiman, Muhammad**; Salman, Mohammed Abdullah. “Cuckoo Search Optimized Fractional Physics-Informed Neural Networks for Fourth-Order Time-Fractional Partial Differential Equations.” *Journal of Mathematics*, 2026. Publisher: Wiley. Reference type: Journal Article.

[101] Saad, Muhammad; **Sulaiman, Muhammad**; Khan, Naveed Ahmad; Laouini, Ghaylen; Alshammari, Fahad Sameer; Rahman, Izaz Ur; Ali, and Mustafa Ahmed. “Neurocomputing Modelling of Hydromagnetic Nanofluid Flow with Chemical Reactions on Fluctuating Sheets.” *Computational and Mathematical Methods*, 2026. Publisher: Wiley. Reference type: Journal Article.

[100] Fazal, Fazl Ullah; **Sulaiman, Muhammad**; Ahmed Ali, Mustafa. “Numerical Simulation of Generalized Fractional Thermoelasticity with Two Temperature and Phase-Lags in a Half-Space using GLA-PINN.” *Complexity*, Vol. 2026, 2026. Publisher: Wiley. Reference type: Journal Article.

[99] Saad, Muhammad; **Sulaiman, Muhammad**; Ur Rahman, Izaz; Alshammari, Fahad Sameer; Laouini, Ghaylen; Abdullah Salman, Mohammed. “Learning PINN-Based Coupled Fractional Systems with Memory and Nonlocal Dynamics Using Neural Computing.” *Journal of Mathematics*, Vol. 2026, 2026. Publisher: Wiley. Reference type: Journal Article.

[98] Jan, Hamad; **Sulaiman, Muhammad**; Fawad Khan, Muhammad; Laouini, Ghaylen; Rahman, Izaz Ur; Abdullah Salman, Mohammed. “Predictive Modeling of Pest Spread in Tea Plants Using an Intelligent Computational Approach.” *Scientific Reports*, 2026. Publisher: Springer Nature. Reference type: Journal Article.

[97] Khan, Hafsa; **Sulaiman, Muhammad**; Laouini, Ghaylen; Ur Rahman, Izaz; Salman Abdullah, Mohammed. “Physics-Informed Neural Networks with Caputo-Fabrizio Derivatives for Nonlinear Fractal-Fractional Delay Equations and Chaotic Systems.” *Scientific Reports*, 2026. Publisher: Springer Nature. Reference type: Journal Article.

[96] Jan, Hamad; **Sulaiman, Muhammad**; Ahmad, Tauqeer; Laouini, Ghaylen; Rahman, Izaz Ur; Abdullah Salman, Mohammed. “Computational modeling of Monte Carlo-enhanced PINNs for fractional order differential models with memory effects.” *Scientific Reports*, 2026. Publisher: Springer Nature. Reference type: Journal Article.

[95] Saad, Muhammad; **Sulaiman, Muhammad**; Alshammari, Fahad Sameer; Rahman, Izaz Ur; Salman, Mohammed Abdullah. “Unveiling Hidden Patterns in Energy Consumption with a Multi-Level Machine Learning Framework.” *Research in Mathematics*, 2026. Publisher: Taylor & Francis. Reference type: Journal Article.

2025

[94] Ahmad, Tauqeer; **Sulaiman, Muhammad**; Bassir, David; Alshammari, Fahad Sameer; Laouini, Ghaylen. “Enhanced Numerical Solutions for Fractional PDEs Using Monte Carlo PINNs Coupled with Cuckoo Search Optimization.” *Fractal and Fractional*, 2025. Publisher: MDPI. URL: <https://www.mdpi.com/2504-3110/9/4/225>. Reference type: Journal Article.

[93] Khan, Muhammad Fawad; Hu, Gang; **Sulaiman, Muhammad**. “Transformer model learning driven analysis of Casson fluid flow influenced by Hall currents and Darcy–Forchheimer effects.” *Physics of Fluids*, Vol. 37, No. 6, 2025. ISSN: 1070-6631. Publisher: AIP Publishing. Reference type: Journal Article.

[92] Jan, Hamad; **Sulaiman, Muhammad**; Saad, Muhammad; Khan, Awais; Salman, Muhammad. “Impact of Individual Participation and Their Health Education on the Growth of the Sportsy Population.” *VFAST Transactions on Mathematics*, Vol. 13, No. 1, pp. 59–83, 2025. ISSN: 2309-0022. Reference type: Journal Article.

[91] Khan, A.; Ahmad, T.; Khan, N. A.; **Sulaiman, Muhammad**. “Predictive Modeling of Parameter Variations in Conveyor Belt Dryers using Machine Learning for Improved Drying Performance.” *VFAST Transactions on Mathematics*, Vol. 13, No. 2, pp. 01–19, 2025. URL: <https://doi.org/10.21015/vtm.v13i2.2105>. Reference type: Journal Article.

2024

[90] **Sulaiman, Muhammad**; Waseem, Muhammad; Ali, Addisu Negash; Laouini, Ghaylen; Alshammari, Fahad Sameer. “Defense strategies for epidemic cyber security threats: modeling and analysis by using a machine learning approach.” *IEEE Access*, Vol. 12, pp. 4958–4984, 2024. ISSN: 2169-3536. Publisher: IEEE. Reference type: Journal Article.

[89] Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Lu, Benzhou. “Predictive insights into nonlinear nanofluid flow in rotating systems: a machine learning approach.” *Engineering with Computers*, 2024. Publisher: Springer. Reference type: Journal Article.

[88] **Sulaiman, Muhammad**; Khalaf, Osamah Ibrahim; Khan, Naveed Ahmad; Alshammari, Fahad Sameer; Algburi, Sameer; Hamam, Habib. “Machine Learning-Based Prediction of Heat Transfer Performance in Annular Fins with Functionally Graded Materials.” *Scientific Reports*, 2024. Publisher: SpringerNature. Reference type: Journal Article.

[87] **Sulaiman, Muhammad**; Salman, Muhammad; Laouini, Ghaylen; Alshammari, Fahad Sameer. “Mathematical models for fluid flow in porous media with machine learning techniques for landfill waste leachate.” *Stochastic Environmental Research and Risk Assessment*, 2024. Publisher: Spring. Reference type: Journal Article.

[86] **Sulaiman, Muhammad**; Khalaf, Osamah Ibrahim; Khan, Naveed Ahmad; Alshammari, Fahad Sameer; Hamam, Habib. “Mathematical modeling and machine learning-based optimization for enhancing biofiltration efficiency of volatile organic compounds.” *Scientific Reports*, Vol. 14, No. 1, Article 16908, 2024. ISSN: 2045-2322. Publisher: Nature Publishing Group UK London. Reference type: Journal Article.

[85] Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Lu, Benzhou. “ANN based optimization of nano-beam oscillations with intermolecular forces and geometric nonlinearity.” *International Journal of Solids and Structures*, Vol. 304, Article 113054, 2024. ISSN: 0020-7683. Publisher: Pergamon. Reference type: Journal Article.

[84] Fazal, Fazlullah; **Sulaiman, Muhammad**; Bassir, David; Alshammari, Fahad Sameer; Laouini, Ghaylen. “Quantitative Analysis of the Fractional Fokker–Planck Levy Equation via a Modified Physics-Informed Neural Network Architecture.” *Fractal and Fractional*, 2024. Publisher: MDPI. URL: <https://www.mdpi.com/2504-3110/8/11/671>. Reference type: Journal Article.

2023

[83] Alshammari, Fahad Sameer; Jan, Hamad; **Sulaiman, Muhammad**; Prathumwan, Din; Laouini, Ghaylen. “Quantitative study of non-linear convection diffusion equations for a rotating-disc electrode.” *Entropy*, Vol. 25, No. 1, Article 134, 2023. ISSN: 1099-4300. Publisher: MDPI. Reference type: Journal Article.

[82] **Sulaiman, Muhammad**; Khan, Naveed Ahmad; Alshammari, Fahad Sameer; Laouini, Ghaylen. “Performance of heat transfer in micropolar fluid with isothermal and isoflux boundary conditions using supervised neural networks.” *Mathematics*, Vol. 11, No. 5, Article 1173, 2023. ISSN: 2227-7390. Publisher: MDPI. Reference type: Journal Article.

[81] Khan, Muhammad Fawad; **Sulaiman, Muhammad**; Ali, Addisu Negash; Laouini, Ghaylen; Alshammari, Fahad Sameer; Khalid, Majdi. “A computational study of magneto-convective heat transfer over inclined surfaces with thermodiffusion.” *IEEE Access*, Vol. 11, pp. 57046–57070, 2023. ISSN: 2169-3536. Publisher: IEEE. Reference type: Journal Article.

[80] **Sulaiman, Muhammad**; Khan, Naveed Ahmad. “Predictive modeling of oil and water saturation during secondary recovery with supervised learning.” *Physics of Fluids*, Vol. 35, No. 6, 2023. ISSN: 1070-6631. Publisher: AIP Publishing. Reference type: Journal Article.

[79] **Sulaiman, Muhammad**; Fazal, Fazlullah; Ali, Addisu Negash; Laouini, Ghaylen; Alshammari, Fahad Sameer; Khalid, Majdi. “A stochastic NARX neural network to investigate the carbon capture in the plantations of forests.” *IEEE Access*, Vol. 11, pp. 74702–74721, 2023. ISSN: 2169-3536. Publisher: IEEE. Reference type: Journal Article.

[78] **Sulaiman, Muhammad**; Khan, Awais; Ali, Addisu Negash; Laouini, Ghaylen; Alshammari, Fahad Sameer. “Quantitative analysis of worm transmission and insider risks in air-gapped networking using a novel machine learning approach.” *IEEE Access*, 2023. Publisher: IEEE. Reference type: Journal Article.

2022

[77] Ahmad Khan, Naveed; **Sulaiman, Muhammad**; Jeza Aljohani, Abdulah; Abu Bakar, Maharani; Miftahuddin, Miftahuddin. “Mathematical models of CBSC over wireless channels and their analysis by using the LeNN-WOA-NM algorithm.” *Engineering Applications of Artificial Intelligence*, 2022. Publisher: Elsevier. Reference type: Journal Article.

[76] Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Tavera Romero, Carlos Andrés; Laouini, Ghaylen; Alshammari, Fahad Sameer. “Study of rolling motion of ships in random beam seas with nonlinear restoring moment and damping effects using neuroevolutionary technique.” *Materials*, Vol. 15, No. 2, Article 674, 2022. ISSN: 1996-1944. Publisher: MDPI. Reference type: Journal Article.

[75] Khan, Naveed Ahmad; **Sulaiman, Muhammad**; Kumam, Poom; Alarfaj, Fawaz Khaled. “Application of Legendre polynomials based neural networks for the analysis of heat and mass transfer of a non-Newtonian fluid in a porous channel.” *Advances in Continuous and Discrete Models*, Vol. 2022, No. 1, pp. 1–32, 2022. Publisher: SpringerOpen. Reference type: Journal Article.

[74] Ahmad Khan, Naveed; **Sulaiman, Muhammad**; Tavera Romero, Carlos Andrés; Alshammari, Fahad Sameer. “Analysis of Nanofluid Particles in a Duct with Thermal Radiation by Using an Efficient Metaheuristic-Driven Approach.” *Nanomaterials*, 2022. Publisher: MDPI. Reference type: Journal Article.

[73] Ahmad Khan, Naveed; **Sulaiman, Muhammad**; Bonya, Ebenezer; Seidu, Jamel; Alshammari, Fahad Sameer. “Investigation of three-dimensional condensation film problem over an inclined rotating disk by using nonlinear autoregressive exogenous model.” *Computational Intelligence and Neuroscience*, 2022. Publisher: Hindawi. Reference type: Journal Article.

[72] Ganie, Abdul Hamid; Fazal, Fazlullah; Tavera Romer, Carlos Andrés; **Sulaiman, Muhammad**. “Quantitative features analysis of water carrying nanoparticles of alumina

over a uniform surface.” *Nanomaterials*, Vol. 12, No. 5, Article 878, 2022. Publisher: MDPI. Reference type: Journal Article.

[71] Guo, Jiayi; Khan, Adnan; **Sulaiman, Muhammad**; Kumam, Poom. “A novel neuroevolutionary paradigm for solving strongly nonlinear singular boundary value problems in physiology.” *IEEE Access*, Vol. 10, pp. 21979–22002, 2022. ISSN: 2169-3536. Publisher: IEEE. Reference type: Journal Article.

[70] Khan, Muhammad Fawad; **Sulaiman, Muhammad**; Romero, Carlos Andres Tavera; Alshammari, Fahad Sameer. “A quantitative study of non-linear convective heat transfer model by novel hybrid heuristic driven neural soft computing.” *IEEE Access*, Vol. 10, pp. 34133–34153, 2022. ISSN: 2169-3536. Publisher: IEEE. Reference type: Journal Article.

[69] **Sulaiman, Muhammad**; Umar, Muhammad; Nonlaopon, Kamsing; Sameer Alshammari, Fahad. “The Quantitative Features Analysis of the Nonlinear Model of Crop Production by Hybrid Soft Computing Paradigm.” *Agronomy*, 2022. Publisher: MDPI. Reference type: Journal Article.

[68] Ganie, Abdul Hamid; Rahman, Irfan Ur; **Sulaiman, Muhammad**; Nonlaopon, Kamsing. “Solution of nonlinear reaction-diffusion model in porous catalysts arising in micro-vessel and soft tissue using a metaheuristic.” *IEEE Access*, Vol. 10, pp. 41813–41827, 2022. ISSN: 2169-3536. Publisher: IEEE. Reference type: Journal Article.

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